

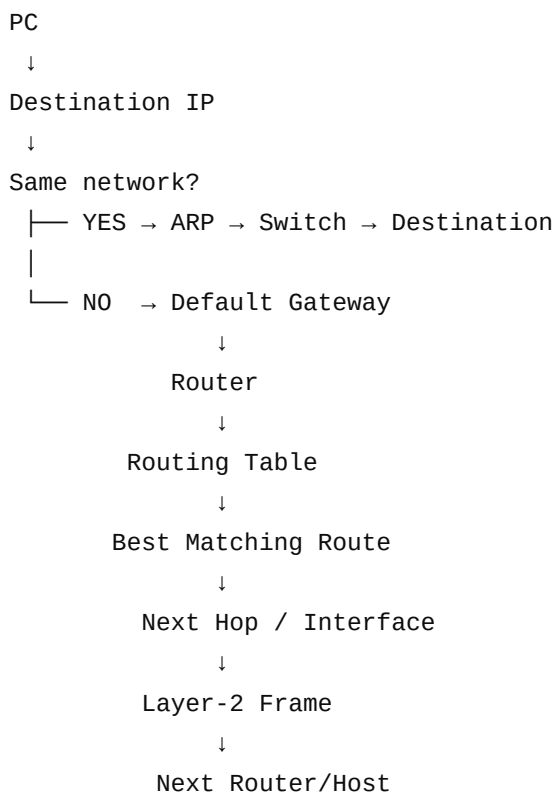
Day 9 — Routing Fundamentals & Routing Tables

Today we'll connect **Day 7 (Layer 2 switching)** and **Day 8 (IP/subnetting)** to understand exactly how a **router decides where to send an IP packet**.

This is very important for an Embedded Linux/networking interview.

Day 9 Goals

By the end of today, you should understand:



You should be able to explain:

- What routing means
- What a routing table is
- Directly connected routes
- Default route
- Next hop
- Gateway
- Interface
- Longest-prefix match
- Static vs dynamic routing
- What happens inside a router
- How Linux performs routing

1. What Is Routing?

Routing = deciding where an IP packet should go next.

Suppose:

PC-A
192.168.1.10

wants to communicate with:

PC-B
192.168.2.20

They are on different networks.

PC-A cannot directly send an Ethernet frame to PC-B.

Instead:

```
PC-A
|
| IP destination = 192.168.2.20
▼
Default Gateway
|
▼
Router
|
▼
192.168.2.0/24
|
▼
PC-B
```

The router performs the **routing decision**.

2. Switching vs Routing

This distinction is extremely important.

Switch

Primarily works at:

Layer 2

It uses:

MAC address

to decide where to forward an Ethernet frame.

MAC



Switch



Port

Router

Primarily works at:

Layer 3

It uses:

Destination IP

to decide where to forward a packet.

Destination IP



Routing table



Next hop/interface

Remember:

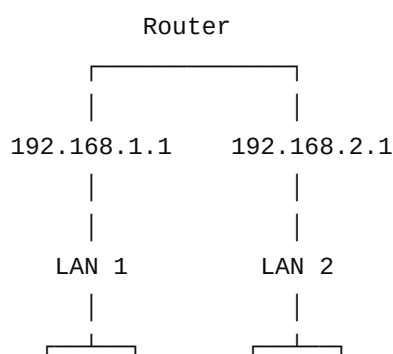
Switch → MAC → Layer 2

Router → IP → Layer 3

A modern device can contain both switching and routing functionality, but this is the conceptual distinction.

3. Simple Network

Consider:





Addresses:

PC-A = 192.168.1.10
PC-C = 192.168.1.20

PC-B = 192.168.2.10
PC-D = 192.168.2.20

The router connects:

192.168.1.0/24
↓
ROUTER
↓
192.168.2.0/24

4. What Is a Routing Table?

A routing table is information that tells the router/kernel:

For this destination network, send the packet through this interface or next hop.

Conceptually:

Destination Network	Next Hop	Interface

192.168.1.0/24	Direct	eth0
192.168.2.0/24	Direct	eth1
0.0.0.0/0	192.168.1.1	eth0

Think of it like a map:

Destination
↓
Where should I send it?
↓
Which interface?
↓
Which next hop?

5. Linux Routing Table

Since you're preparing for a Linux networking interview, this is important.

Run:

```
ip route
```

You might see:

```
default via 192.168.1.1 dev eth0  
192.168.1.0/24 dev eth0 proto kernel scope link src 192.168.1.10
```

Let's understand it.

6. 192.168.1.0/24 dev eth0

```
192.168.1.0/24 dev eth0
```

means approximately:

The 192.168.1.0/24 network is directly reachable through eth0 .

Visual:

```
Linux PC  
192.168.1.10  
  |  
  | eth0  
  ▼  
192.168.1.0/24
```

No router is needed for another host on that same subnet.

7. default via

Example:

```
default via 192.168.1.1 dev eth0
```

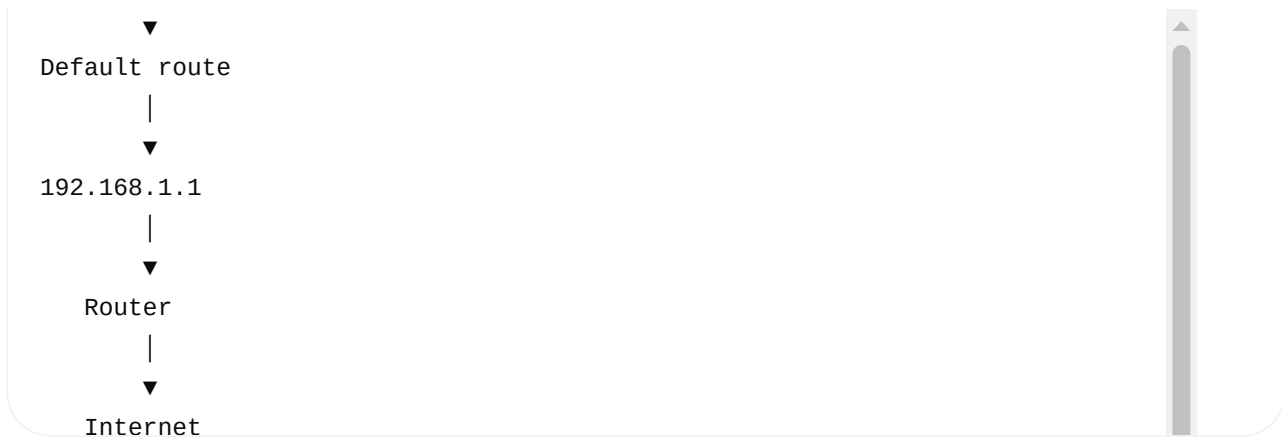
means:

If no more specific route matches, send the packet to 192.168.1.1 through eth0 .

Visual:

```
Unknown destination  
  |
```





8. What Is a Default Route?

The default route is:

0.0.0.0/0

Why?

Because /0 means:

0 network bits

So it matches essentially every IPv4 destination.

But it is the **least specific** route.

That's why:

The default route is used only when a more specific matching route isn't available.

9. Example Routing Table

Suppose:

Destination	Gateway	Interface

192.168.1.0/24	Direct	eth0
192.168.2.0/24	Direct	eth1
10.10.0.0/16	192.168.1.254	eth0
0.0.0.0/0	192.168.1.1	eth0

Now a packet arrives with:

Destination = 192.168.2.50

Router searches the table.

192.168.1.0/24	✗
192.168.2.0/24	✓
10.10.0.0/16	✗
default	✓

There are two conceptual matches only if the default is considered; the more specific route wins:

192.168.2.0/24

Therefore:

Use eth1

10. Longest Prefix Match

This is one of the **most important Day 9 interview concepts**.

Suppose the routing table contains:

10.0.0.0/8
10.10.0.0/16
10.10.20.0/24
default

Destination:

10.10.20.50

It matches:

10.0.0.0/8
10.10.0.0/16
10.10.20.0/24
0.0.0.0/0

Which one wins?

10.10.20.0/24

Because:

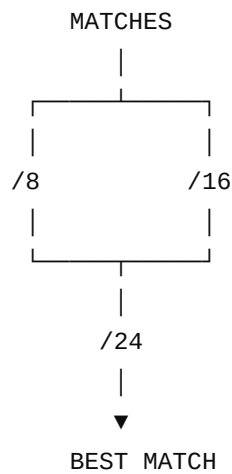
/24 > /16 > /8 > /0

This is **longest-prefix match**.

11. Visualize Longest Prefix Match

Destination:

10.10.20.50



Think:

More prefix bits = more specific route.

12. What Is a Next Hop?

Suppose router has:

Destination:

10.10.20.0/24

Next Hop:

192.168.1.254

It means:

Send the packet to the router at 192.168.1.254 next.

Visual:

Router-A



| packet for 10.10.20.x

Next-hop Router

192.168.1.254



Other network

Important:

Next hop is not necessarily the final destination.

13. Destination vs Next Hop

Suppose:

PC-A

192.168.1.10

wants:

Server

10.10.20.50

Routing table says:

10.10.20.0/24

via 192.168.1.254

Then:

Destination IP = 10.10.20.50

Next Hop = 192.168.1.254

These are different.

Destination



Final IP

Next Hop



Immediate router/interface to which
the packet is forwarded

14. What Does the Router Actually Receive?

Suppose PC-A sends:

Destination IP = 10.10.20.50

Ethernet frame arrives at router.

Conceptually:

Ethernet Frame

Destination MAC = Router
Source MAC = PC-A
IP Packet
Src IP = 192.168.1.10
Dst IP = 10.10.20.50

The router:

1. Receives Ethernet frame
- ↓
2. Removes/decapsulates Layer 2 header
- ↓
3. Looks at destination IP
- ↓
4. Searches routing table
- ↓
5. Selects best route
- ↓
6. Determines outgoing interface/next hop
- ↓
7. Builds a new Layer-2 frame
- ↓
8. Transmits through outgoing interface

15. Very Important: Router Rebuilds Layer 2

This connects directly to your previous question.

Suppose:

PC-A → Router → PC-B

First frame:

PC-A → Router

MAC:

Source MAC = MAC-A
Destination MAC = MAC-R1

After router processing:

Router → PC-B

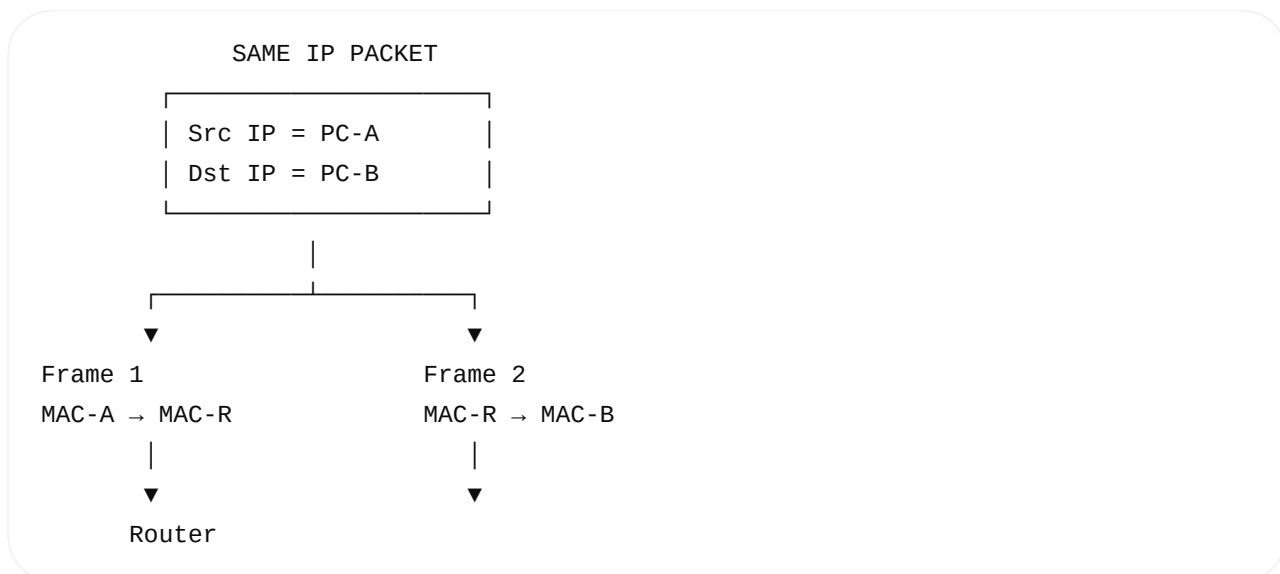
New Ethernet frame:

Source MAC = MAC-R2
Destination MAC = MAC-B

But IP packet still targets:

Destination IP = PC-B

Visual:



Layer 2 changes at each routed hop.

16. What About the TTL?

There's another important Layer 3 operation performed by routers.

IPv4 packets contain:

TTL

TTL = Time To Live

Every router forwarding an IPv4 packet decrements the TTL.

Example:

```
PC-A
TTL = 64
|
▼
Router 1
TTL = 63
|
▼
Router 2
TTL = 62
|
▼
Router 3
TTL = 61
```

Why?

To prevent packets from circulating forever because of routing loops.

17. What Happens When TTL Reaches Zero?

Suppose:

```
TTL = 1
```

Router receives it.

It decrements:

```
1 → 0
```

The packet is discarded.

The router can send an:

```
ICMP Time Exceeded
```

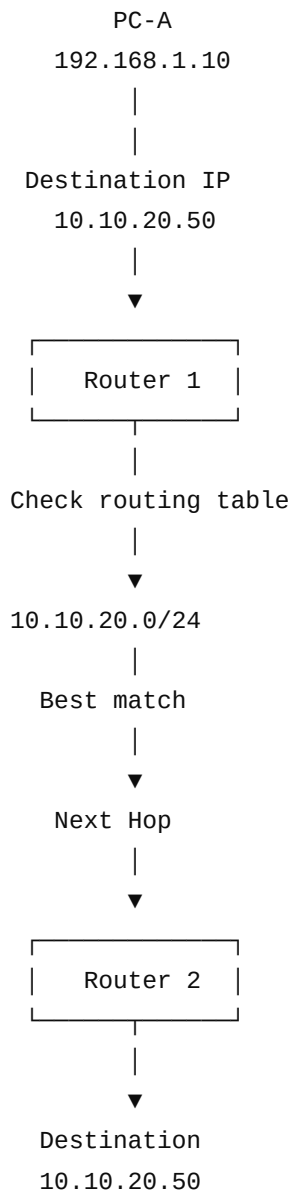
message back toward the source.

This is exactly related to how **traceroute** works.

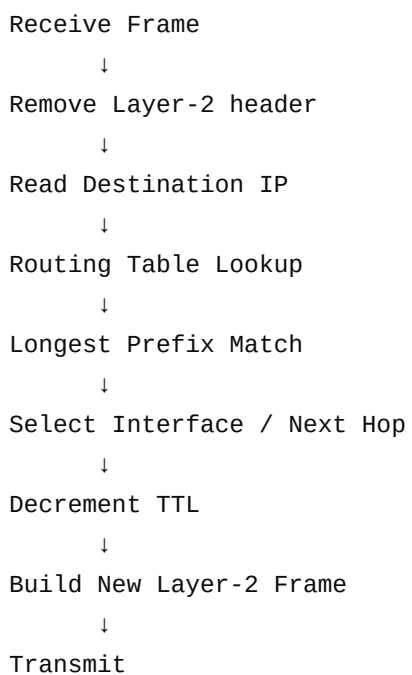
We'll study ICMP and traceroute on **Day 10**.

18. Routing Process — Full Visualization

Now combine everything:



At each router:



19. Directly Connected Route

Suppose Linux machine has:

eth0 = 192.168.1.10/24

Linux automatically knows:

192.168.1.0/24

is directly connected.

So:

192.168.1.50

doesn't need the default gateway.

Flow:

```
Application
  ↓
Destination = 192.168.1.50
  ↓
Routing lookup
  ↓
192.168.1.0/24
  ↓
eth0
  ↓
ARP for 192.168.1.50
  ↓
MAC address
  ↓
Ethernet
  ↓
PC-B
```

20. Different Network

Now destination:

192.168.2.50

Routing table:

```
192.168.1.0/24 dev eth0
default via 192.168.1.1 dev eth0
```

No specific route for:

```
192.168.2.0/24
```

Therefore:

```
Destination
192.168.2.50
  ↓
No specific route
  ↓
Default route
  ↓
192.168.1.1
  ↓
Gateway
```

21. ARP's Role in Routing

This is another important connection between **Layer 2 and Layer 3**.

Suppose:

```
PC-A = 192.168.1.10
Gateway = 192.168.1.1
Destination = 192.168.2.50
```

PC-A does **not** ARP for:

```
192.168.2.50
```

because the destination is on another network.

Instead:

```
PC-A
  |
  | "Who has 192.168.1.1?"
  ▼
ARP
  |
  ▼
Gateway MAC
```

Then:

Ethernet:
Destination MAC = Gateway MAC

IP:
Destination IP = 192.168.2.50

This distinction is extremely important.

22. Remember This Rule

Same network

ARP → Destination IP

Example:

ARP for 192.168.1.20

Different network

ARP → Default Gateway IP

Example:

ARP for 192.168.1.1

The IP destination remains:

192.168.2.20

23. Static Routing

A **static route** is manually configured.

Example Linux command:

```
sudo ip route add 10.10.20.0/24 via 192.168.1.254
```

Meaning:

Destination:
10.10.20.0/24



Next hop:
192.168.1.254

Conceptually:

10.10.20.0/24
|
▼
192.168.1.254

This is useful for small networks or specific controlled routes.

24. Dynamic Routing

In larger networks, manually configuring every route becomes difficult.

Routers can use routing protocols to learn routes.

Examples:

RIP
OSPF
BGP
IS-IS

Conceptually:

Router A
|
| routing protocol
▼
Router B
|
| learns networks
▼
Routing Table

You don't need to deeply study all of these yet.

For your current interview preparation, understand:

Static routing
vs
Dynamic routing

25. Static vs Dynamic

Feature	Static	Dynamic
Configuration	Manual	Learned
Complexity	Simple for small networks	Better for larger networks
Adaptation to failures	Limited/manual	Can adapt
Overhead	Low	Routing protocol overhead
Example	ip route add	OSPF/BGP/RIP

26. Linux: View Routing Table

Use:

```
ip route
```

You can also use:

```
ip route show
```

For a specific destination:

```
ip route get 8.8.8.8
```

This is **very useful for interviews and debugging**.

Example:

```
$ ip route get 8.8.8.8
8.8.8.8 via 192.168.1.1 dev eth0
```

Conceptually Linux is telling you:

```
Destination = 8.8.8.8
      ↓
Next hop = 192.168.1.1
```



27. Linux Networking Stack Connection

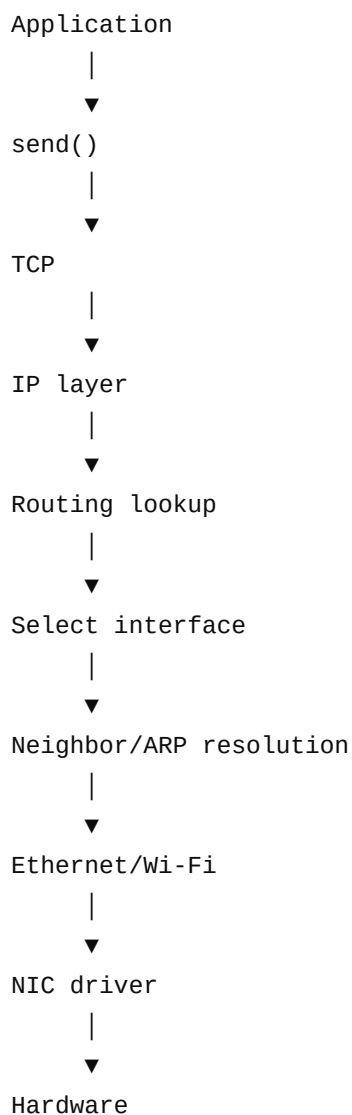
You previously asked where `send()` , `connect()` , etc. interact with the layers.

Now connect routing to that.

Suppose:

```
send(sock, data, len, 0);
```

Conceptually:



So routing is part of the **kernel's Layer 3 processing**.

28. Does `connect()` Directly Call the Router?

No.

This is an important correction to the concept we discussed earlier.

When an application calls:

```
connect()
```

the application is asking the socket layer to establish/connect the transport endpoint.

For TCP:

```
connect()
  ↓
TCP connection processing
  ↓
SYN
  ↓
IP routing
  ↓
Layer 2
  ↓
NIC
```

The router is not directly called by your application.

The kernel's networking stack determines whether the packet needs to go to:

```
Local destination
```

or:

```
Gateway / router
```

based on the routing decision.

29. Complete Example

Let's put everything together.

PC-A:

```
IP:
192.168.1.10/24
```

Gateway:

```
192.168.1.1
```

Server:

```
10.10.20.50
```

Application calls:

```
connect(10.10.20.50, 8080)
```

Step 1 — Application

```
connect()
```

Step 2 — TCP

TCP creates:

```
Source Port = ephemeral port  
Destination Port = 8080
```

and begins connection establishment.

Step 3 — IP

Destination:

```
10.10.20.50
```

Step 4 — Routing

Linux checks:

```
192.168.1.0/24  
default via 192.168.1.1
```

Destination isn't in:

```
192.168.1.0/24
```

Therefore:

```
Use default route
```

Step 5 — ARP

Linux needs the gateway's MAC:

Who has 192.168.1.1?

Step 6 — Ethernet

Src MAC = PC-A

Dst MAC = Router

Step 7 — Router

Router receives frame.

Remove Layer-2 header



Read destination IP



10.10.20.50



Routing table



Choose next hop

Step 8 — New Ethernet Frame

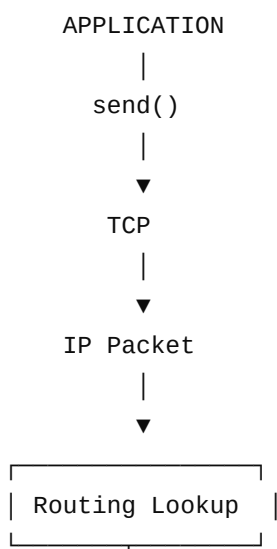
Router creates another Layer-2 frame for the next link.

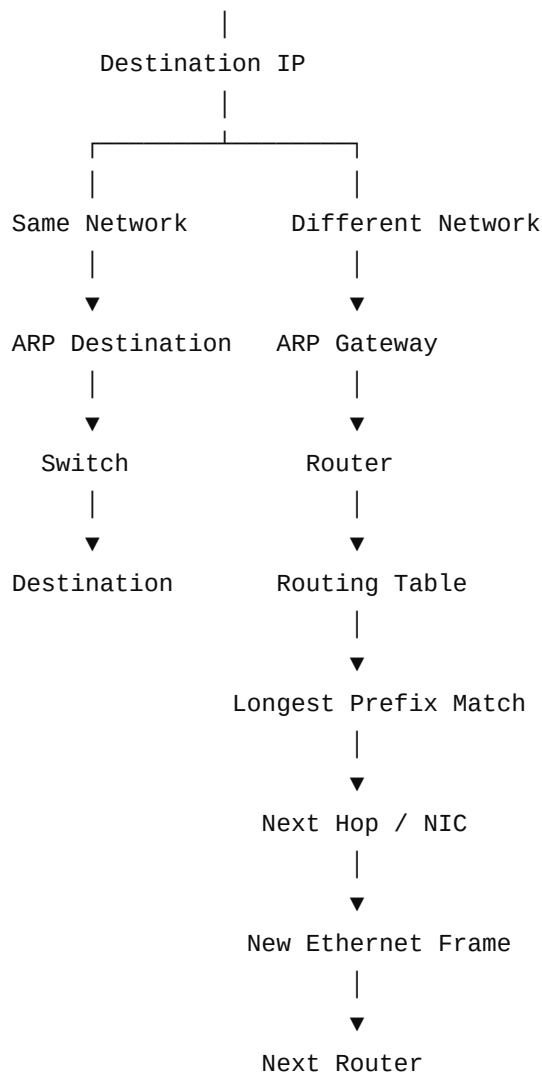
Router MAC → Next Hop MAC

And this repeats until the destination network is reached.

30. The Most Important Diagram of Day 9

Keep this one for revision:





★ Day 9 Interview Must-Know

1. What is routing?

Routing is the process of determining the path/next hop for an IP packet based on its destination IP and the routing table.

2. What does a router primarily use?

Destination IP address and its routing table.

3. What does a switch primarily use?

Destination MAC address and its MAC address table.

4. What is a default route?

0.0.0.0/0

It is used when no more specific route matches.

5. What is longest-prefix match?

When multiple routes match a destination, the most specific route—the one with the longest prefix—is selected.

6. What is next hop?

The immediate router/device to which a packet is forwarded on its way to the final destination.

7. Does the MAC address remain unchanged across routers?

No.

MAC → changes at each Layer-2 hop
IP → normally remains end-to-end

8. What happens to IPv4 TTL?

Each router:
 $TTL = TTL - 1$

Day 9 Practice

Try to solve these **without looking at the answers**.

Question 1

Routing table:

192.168.1.0/24 dev eth0
10.0.0.0/8 via 192.168.1.1
0.0.0.0/0 via 192.168.1.254

Destination:

10.20.30.40

Which route is selected?

Question 2

Routing table:

10.0.0.0/8
10.10.0.0/16
10.10.20.0/24
0.0.0.0/0

Destination:

10.10.20.100

Which route wins?

Question 3

PC-A:

192.168.1.10/24

Destination:

192.168.1.50

Does PC-A ARP for the destination or the gateway?

Question 4

PC-A:

192.168.1.10/24

Destination:

192.168.2.50

Does PC-A ARP for 192.168.2.50 or its gateway?

Question 5

A packet crosses three routers.

Initial TTL:

64

What is the TTL after the third router forwards it?

Answers

Q1 → 10.0.0.0/8

Q2 → 10.10.20.0/24

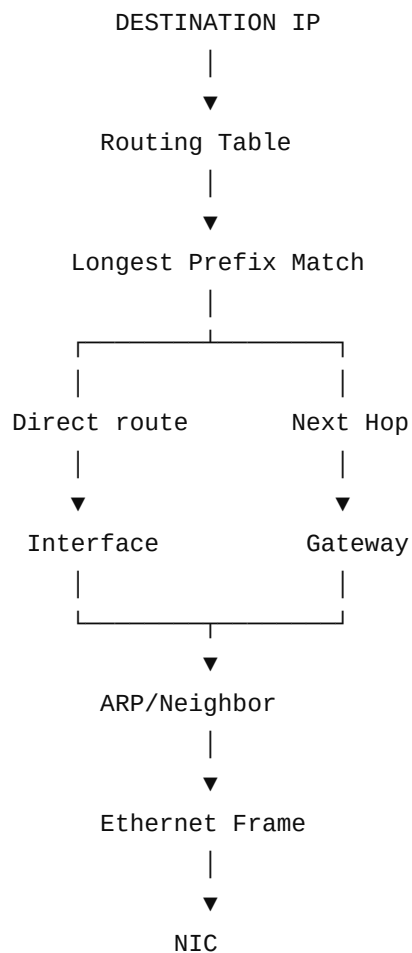
Q3 → ARP for 192.168.1.50

Q4 → ARP for the default gateway

Q5 → 61

Day 9 Final Mental Model

If you remember only this, you're in good shape:



Day 9 is complete. Next in the fixed schedule is Day 10: ICMP, ping , traceroute , and how Linux discovers where a network path is failing.